MILLIMETER-WAVE IAB ENABLED MOBILE EDGE COMPUTING WIRELESS NETWORKS

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Abstract: This study explores the implementation of a hybrid routing and computation offloading framework, based on multi-objective optimisation for a 5G mmWave Integrated Access and Backhaul (IAB) network enabled with Multi-Access Edge Computing (MEC). The framework aims to minimize end-to-end latency and user equipment energy consumption while maintaining service continuity under dynamic network conditions. Unlike conventional IAB-MEC models that assume static links and uniform nodes, the proposed framework incorporates time varying link state awareness and a multihop energy penalty model, enabling routing and offloading decisions to adapt to real time propagation changes and cumulative hop costs. The routing component leverages a breadth first search algorithm and a score-based path evaluation scheme to prevent IAB node congestion, while the offloading model applies a slice specific weighted sum optimisation approach that considers task priority, queue delay and server load capacity to meet service deadlines. The offloading algorithm is evaluated in ns-3 using the mmWave module, which implements the 3GPP protocol stack to ensure accurate results. Data is exported as CSV files and visualized in MATLAB for performance analysis. Results indicate that the proposed algorithm maintains a balanced trade-off between energy efficiency (89.3 %) and offloading rate (43.9 %) while achieving minimal latency, full deadline adherence, and 100 % reliability across multiple network slices. The results indicate the frameworks suitability for next generation IAB based MEC networks in supporting real time services.

Key words: 5G mmWave, Computation Offloading, Integrated Access and Backhaul, Multi-Access Edge Computing

1. INTRODUCTION

The rapid expansion of technological systems towards an integrated society through smart cities, artificial intelligence, autonomous driving and virtual reality pushed the need for a faster, more reliable and ultra-low latency communication system. This led to the development of the fifth generation (5G) network [1], offering data rates up to 20 Gbps, latency as low as 1 ms and support for up to a thousand times more connected devices than 4G [2]. Such performance can only be achieved by signals operating at frequencies above 30 GHz, referred to as millimetre waves (mmWaves), due to their short wavelength. However, the short wavelength of mmWave signals leads to high pathloss and signal reflection from solid objects. While directional beamforming can mitigate these effects by focusing the signal energy towards specific users, it introduces additional challenges such as the need for continuous beam tracking, which increases system overhead and may degrade throughput [3].

To address these limitations, a high-density deployment of next generation node bases (gNB) can be employed to enhance network capacity, coverage and quality of service (QoS) by reducing propagation losses and ensuring more consistent line of sight (LOS) link conditions. Such deployments however, cannot rely on wired backhauling for every gNB due to high operational and capital expenditure for network providers. To make such dense deployment economically viable, 3GPP standardised the Integrated Access and Backhaul (IAB) architecture [4], which employs mmWave wireless backhauling to interconnect base stations via multi-hop links delivering comparable performance and connectivity at a lower cost. While IAB networks significantly enhance coverage and connectivity their increased communication capabilities raise user equipment (UE) energy consumption. This occurs as users and internet of things (IoT) devices leverage the improved network to perform computationally intensive tasks such as virtual reality, remote surgery, automated manufacturing and autonomous driving often surpassing the devices optimal energy and performance balance. To mitigate the energy efficiency concerns while maintaining the QoS required for latency sensitive applications, the integration of Mobile Edge Computing (MEC) within 5G networks has emerged as a key solution. By bringing computing resources closer to the network edge, MEC enables the offloading of demanding computations to nearby edge servers, thereby reducing latency, improving user experience and enhancing UE energy efficiency [5]. This paper proposes a multi-objective optimisation algorithm to minimise UE energy efficiency while meeting QoS latency constraints across different 5G network slices (URLLC, eMMB, mMTC).

The rest of this paper is structured as follows. Section 2 reviews existing computation offloading strategies and identifies the research gap. Section 3 presents the network topology model, assumptions, constraints and mathematical formulations. Section 4 addresses the ns-3 simulation results and benchmarking against other computation offloading methods. Section 5 concludes with future considerations.

2. BACKGROUND

2.1 Literature Review

The integration of 5G and edge computing has inspired numerous offloading strategies, aimed at optimising latency, energy consumption and network throughput. Machine learning approaches such as deep learning (DL), reinforcement learning (RL), and deep reinforcement learning (DRL) enable adaptive computation offloading without explicit programming. Dong *et al* [6], showed that DL can minimise energy use by optimising user association and offloading probabilities while maintaining QoS. Multi-agent DRL models [7], [8] extend learning to distributed agents, improving coordination and scalability. For example Gan *et al* [7] achieved a 5.4% reduction in execution latency compared to traditional algorithms such as Deep Deterministic Policy Gradient (DDPG) and Deep Q-Learning (DQN). Similarly, Zhai *et al* [8] proposed a Dueling-DQN algorithm that produced a 13.4% improvement in response quality on edge servers and a 5% reduction in total response time over Q-learning and cache replacement policies (LRU, LFU) showing the ability of a multi-agent approach to minimising delay and maximising resource allocation. Joint optimisation techniques [9], [10] address multiple interdependent network parameters such as power and bandwidth allocation, often forming NP-hard problems. Sardellitti *et al* [9], jointly optimised radio and computation resources reducing energy use by 45.8 % while Zhang *et al* [10] achieved an 18 % energy reduction across 100 devices using their ECCO scheme. Such NP-hard problems are frequently simplified using game theory, which models optimal strategies among multiple users, as shown by Yuan *et al* [11], whose game theoretic offloading strategy achieved an average task delay of 40ms for a 30 UE network. When balancing conflicting objectives, multi-objective optimisation offers effective trade-offs. Alfakih *et al* [12], used an accelerated particle swarm algorithm based on multi-objective optimisation and managed to reduce task completion time by 30% and improve resource allocation by 29 %. Despite their low complexity, simpler heuristic algorithms can yield performance with less overhead. Barbosa de Souza *et al* [13], used a greedy algorithm achieving a 71.8% offloading success rate and a 54.1 % reduction in offloading time, while Yang *et al* [14] applied an artificial fish swarm algorithm for 10 % energy savings. These results show that lightweight designs can provide efficient offloading for 5G networks at reduced complexity.

While optimisation strategies have been explored in 5G NOMA [15], Hentet [10] and V2X [13], computation offloading in 5G mmWave IAB systems with realistic propagating dynamics remain unexplored. Prior studies neglect the effects of shared backhaul bandwidth and multi-hop relays on energy efficiency and real time performance. This paper seeks to fill this gap by analysing computation offloading within a 5G mmWave IAB enabled network.

3. SYSTEM OVERVIEW

3.1 Network Topology

The network topology consists of a central gNB, shown by the red dot in figure 1. It is collocated with a MEC server that functions as a computational hub and a network root. It is interconnected to 8 IAB nodes strategically positioned to model urban densification in an urban environment. The topology forms a directed acyclic graph (DAG), compliant with the 3GPP IAB framework [16], preventing routing loops and ensuring efficient data flow from UEs towards the donor gNB. The network operated under a set of practical constraints, with each IAB node supporting a maximum of four simultaneous connections to reflect practical capacity constraints and maintain manageable interference levels. The MEC server has a fixed computational capacity of 20 concurrent tasks, and a queue size limit of 10, with a 20 % drop in CPU processing efficiency at full load to emulate realistic bottlenecking. The maximum path length between any UE and the donor is limited to 3 hops to mitigate compounding latency, reliability loss and bandwidth degradation across multiple hops. The urban environment introduces highly dynamic signal propagation, as backhaul links between UEs and IAB nodes are subject to randomised, time varying line of sight (LOS), non-line of sight (NLOS) and blockage conditions. This enables the simulation of obstacles such as buildings, vegetation, heavy rainfall and equipment breakdowns all of which severely impact 5G mmWave IAB performance [1].These conditions are modelled using the 3GPP Urban Microcell (UMi) Street Canyon propagation model in ns-3, as link states vary between LOS, NLOS and blocked conditions.

Due to multiple potential paths and dynamic link states, an intelligent routing mechanism is required [16]. Since 3GPP does not specify a route selection policy for complex IAB networks [17], this work adopts a Breath First Search (BFS) algorithm for route selection due to its low time complexity [18] providing an efficient and scalable means of identifying candidate paths. To simulate a realistic network load user task generation follows a Poisson distribution model. This stochastic model accurately represents the random and independent nature of task arrival from multiple UEs in a live network, allowing for assertive algorithm performance metrics. The system incorporates three priority classes of computation tasks with varying size, complexity and latency requirement across different UE groups to model heterogenous application demands. The system level parameters are derived from [4] and summarised in Appendix C.

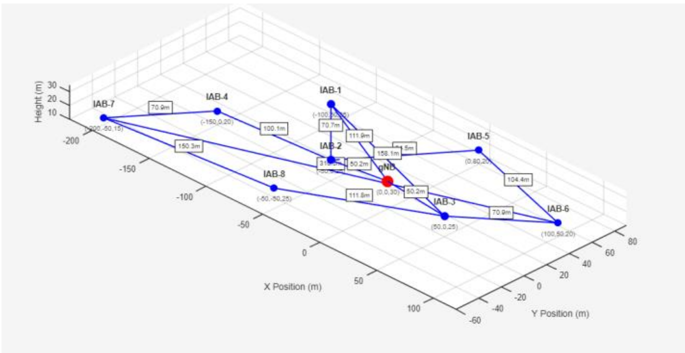

Figure 1: Topology of the proposed 5G IAB network

3.2 Mathematical Formulation

In the proposed model, each UE generates tasks to be considered for local execution or computational offloading. Each generated task has the following properties, S_i denoting the task size, C_i denoting the required CPU cycles, D_i denoting the task deadline for processing, O_i denoting the output data and priority mapping to any of the network slices (URLLC, eMMB and mMTC). A detailed list on the classification of data packet sizes per network slice is provided in Appendix D. Assuming that the generated task computation is performed locally, the latency for local computation is calculated by the equation:

$$T_{local} = \frac{C_i}{f_{UE}}, \tag{1}$$

where C_i are the CPU cycles required by the task and f_{UE} is the UE CPU processing speed. The resulting energy consumption for a locally processed task is calculated using the standard power model for CMOS circuits [11]:

$$E_{local} = \kappa * f_{UE}^2 * C_i\,, \tag{2}$$

where κ is the coefficient related to energy consumption. The end-to-end latency for offloading a task is a sum of several individual delays, namely uplink, queueing, processing and downlink. The time to send the task over N hops to the MEC server is calculated by the equation:

$$T_{uplink} = \sum_{i=0}^{N} \left(\frac{S_i}{R_i} + \alpha\right), \tag{3}$$

where R_i is the instantaneous data rate for *i* hops between IAB nodes and α is the constant processing delay per hop. The instantaneous data rate is a measure of the amount of data transported from one device to another per each hop. This is characterised by the Shannon-Hartley equation:

$$R_i = B * \eta * \log_2(1 + SINR), \tag{4}$$

where B is the network bandwidth, η the efficiency factor, $SINR$ is the signal-to-interference-plus-noise ratio at each hope. The efficiency factor accounts for modulation schemes and practical implementation losses that reduce the achievable data rate, thereby influencing downlink and uplink speeds. The SINR at each hop is characterised by the following equation:

$$SINR = P_{TX} + G_{TX} + G_{RX} - PL_i(d_i, S_i) - P_{noise}, \tag{5}$$

where P_{TX} is the transmission power of the source, G_{TX} is the antenna gain of the transmitter, $PL_i(d_i, S_i)$ the path loss signal attenuation which depends on distance and link state (LOS, NLOS, blocked) based on the UMi pathloss model, G_{RX} is the antenna receiving gain and P_{noise} the noise power sum of thermal and noise figures listed in Appendix C. The time the task spends waiting in the MEC server queue when the server is at full capacity is defined by the following equation:

$$T_{queue} = Q_{length} * T_{avg\,processing}\,, \tag{6}$$

where Q_{length} is the number of tasks already in the queue and $T_{avg\,processing}$ is the estimated average processing time for a task. The task processing time can be calculated by the equation:

$$T_{processing} = \frac{C_i}{\beta * f_{MEC}}\,, \tag{7}$$

where f_{MEC} is the MEC server CPU processing speed and β is the CPU processing efficiency factor that adjusts computational efficiency based on server capacity. The time to send the task result back to the UE over the same multi-hop path is calculated by the following equation:

$$T_{downlink} = \sum_{i=0}^{N} \left(\frac{O_i}{R_i} + \delta\right), \tag{8}$$

where O_i is the size of the processed task and δ is the processing delay per hop. The energy consumed by the UE's transmitter to send the task data is estimated by the following equation:

$$E_{offload} = P_{UE} * \frac{T_{upload}}{1000} * (\,1 + (0.1 * N)\,)\,, \tag{9}$$

where P_{UE} is the UE transmission power. This energy model incorporates a 10% per-hop penalty factor expressed as $\big(1 + (0.1 * N)\big)$ to account for the cumulative signal quality degradation across multiple hops, which increases transmission energy consumption. The values of the parameters in equations (1) – (9) are listed in Appendix E.

3.3 Multi-Objective Optimisation

The core optimisation problem is to determine the optimal offloading decision x_i for each task *i* where:

$$x_i = \begin{cases} 1 & if\ the\ task\ is\ offloaded \\ 0 & if\ the\ task\ is\ executed\ locally \end{cases} \tag{10}$$

The objective is to minimise a weighted sum of total energy consumption and end-to-end latency, where the weights are determined by task priority through the equation:

$$\min_{x_i\{0,1\}} \sum [\,\gamma_i E_i(x_i) + \rho_i\, T_i\,(x_i)\,]\,, \tag{11}$$

$$E_i(x_i) = x_i E_i^{offlaad} + (\,1 - x_i)E_i^{local}\,, \tag{12}$$

$$T_i(x_i) = x_i T_i^{offload} + (1 - x_i) T_i^{local}\,. \tag{13}$$

Here γ_i and ρ_i are slice specific weighting coefficients controlling the trade-off between energy efficiency and latency. A higher γ_i emphasises energy savings favouring local execution, while a higher ρ_i prioritises delay reduction encouraging offloading to the edge server. The slice-based optimisation weights are listed in Appendix F.

3.4 Proposed Offloading Algorithm

The computational offloading algorithm functions as an adaptive decision-making framework that dynamically balances energy efficiency, latency, and network load. Once a task is generated, the routing policy identifies all feasible paths between the user and the MEC server, selecting the most efficient route based on current network conditions. Candidate paths are evaluated by hop count and IAB node utilisation, with preference given to the least congested and unblocked route. When link blockages occur, the algorithm automatically reroutes traffic to alternative gNB paths to preserve service continuity. These routing updates are then integrated into the offloading decision process, which allocates tasks according to the energy and latency trade-offs defined in equations (11) – (13). The system ultimately determines whether each task should be executed locally or offloaded to the MEC server on the basis of the cost to offload providing sufficient benefit compared to local execution. Pseudocode implementations of the routing, link-state, and offloading algorithms are provided in Appendices G1–G3.

4. SIMULATION RESULTS AND ANALYSIS

4.1 Latency

As shown by figure 2, the proposed optimisation algorithm achieves a maximum latency of 17.7 ms, a minimum latency of 4.4 ms and an average of 11.5 ms across 316 URLLC tasks. Within the URLLC slice, an average network throughput of 3.25 Gbps, peaking at 5.42 Gbps is also achieved, indicating that latency improvements are met without compromising data rate performance. Overall, the system remains well within the 1-30 ms latency and 1 – 100 Mbps data rate requirements for instantaneous control and safety critical applications in industrial automation and smart grids [19]. The achieved latency range also supports medium complexity tasks such as industrial augmented reality (AR), which typically requires latency below 50 ms for seamless rendering. Moreover, the algorithm demonstrates sufficient scalability for large scale workloads, including high definition (HD) video analytics and real time digital twin simulations which can tolerate up to 100 ms latency under high throughput conditions.

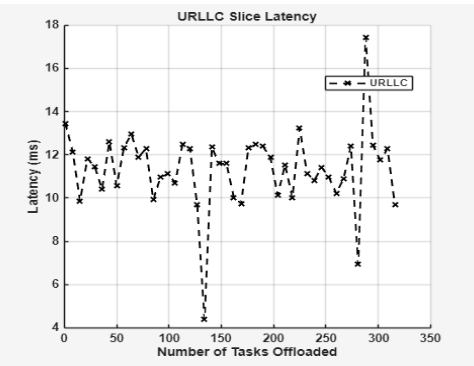

Figure 2: Latency results for the URLLC slice

As illustrated by figure 3, the eMMB network slice achieves a maximum latency of 488.3 ms, a minimum latency of 4.1 ms, and an average latency of 153.4 ms across 2010 tasks. The slice also attains an average network data rate of 3.2 Gbps, peaking at 5.54 Gbps. These results confirm the slices suitability for eMMB services, where high data throughput up to 20 Gbps [2], is prioritised over ultra-low latency. The achieved latency range comfortably supports bandwidth intensive and delay tolerant applications such as HD and ultra-HD video streaming and immersive virtual reality (VR) experiences, which can tolerate up to 200 ms. The proposed algorithm also maintained 100% deadline adherence and reliability for eMBB and URLLC slices in relation to latency, as shown by Appendix H. Highlighting its capability to sustain stringent QoS requirements of 99.999 % reliability for 5G standards [2]. Within the mMTC network slice, no tasks were offloaded. This is primarily because devices in this category such as smart lighting, parking and metering systems generate small, low complexity tasks and operate infrequently to conserve energy allowing lifetimes of up to 10 years on a single battery, hence no energy usage to optimise. As shown by figure 4, which compares local execution and offloading for all mMTC tasks the latency improvement achieved through offloading is negligible. Since the algorithm employs a threshold-based scoring system to confirm the gain achieved by offloading a task. This lack of benefit prevents the offloading score from surpassing the decision threshold resulting in all mMTC tasks being executed locally.

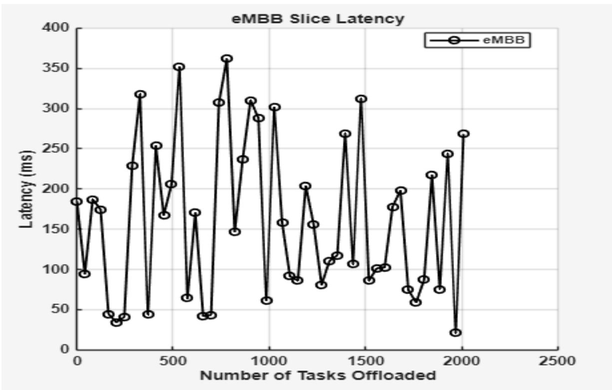

Figure 3: Latency results for the eMMB slice

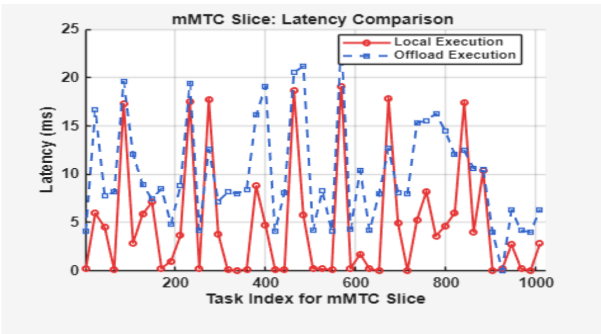

Figure 4: Latency results for the mMTC slice

4.2 UE Energy Efficiency

The energy efficiency of the proposed algorithm was benchmarked against three existing optimisation algorithms GTCOS[11], JOBCA[20] and PTAS-MAMEC [21], using the same network topology described in section 3.1 and the parameter values listed in Appendix C and E, for a fair and reliable performance analysis. GTCOS employs a game theoretic convex optimisation model for resource allocation and energy minimisation. PTAS-MAMEC applies a single objective optimisation for Pareto optimal energy optimisation and JOBCA jointly optimises computation offloading and resource allocation. These algorithms were selected as representative baselines of established offloading optimisation methods. The comparative performance results are summarised in table1.

Table 1: Algorithm performance benchmark

	Energy Efficiency %	Offload Rate %	Time Complexity (On)
Proposed Algorithm	89.3	43.9	$O(n * P^2)$
GTCOS	95.1	30.2	$O(n * P^2)$
JOBCA	59	14.9	$O(n * P^2)$
PTAS-MAMEC	91.6	45	$O(T^q(K - q^3)$

The results in table 1 show distinct performance trends among the evaluated algorithms under the proposed IAB network topology. The proposed algorithm achieves a balanced trade-off between energy efficiency (89.3 %) and the offloading rate (43.9 %), outperforming the JOBCA algorithm in adaptability to the IAB network. The GTCOS algorithm achieved the highest energy efficiency at 95.1 %, but a low moderate offloading rate of 30.2 %. This performance can be attributed to its convex optimisation model, which prioritises energy savings through selective task offloading resulting in reduced overall task distribution across the network. This behaviour is reflected in the uneven per-slice offloading rates, where the algorithm attains 100 % offloading for URLLC tasks but only a 2.5 % offloading rate for the eMBB slice as shown by Appendix I. In contrast the JOBCA algorithm recorded the lowest energy efficiency and offloading rates. This limitation stems from its joint optimisation framework, originally designed for a small cell with a direct backhaul link to a single gNB donor. As a result, the algorithm struggles to adapt to the multi-hop topology and shared backhaul configuration proposed in this study. This outcome reinforces the need for computation offloading algorithms that are specifically designed for the unique characteristics and constraints of IAB networks. The JOBCA algorithm despite its lower efficiency, demonstrates the most balanced offloading rates across network slices with a 13.3 % for URLLC, 20.2 % for eMMB and a 3.6 % for mMTC. Although relatively low as a result of the efficiency rating, these values indicate an even distribution of offloaded tasks across the network compared to other approaches. This suggests that its resource allocation approach could be integrated into the proposed framework to improve slice balance. As the proposed algorithm achieves a 21 % URLLC offload rate and a 72.2 % eMMB offload rate, indicating that many latencies critical URLLC tasks are computed locally. Increasing URLLC offloading may further enhance system responsiveness and increase energy efficiency. The PTAS-MAMEC algorithm, achieves the highest efficiency rating at 91.6 % and the highest offloading rate at 45 %, as a result of the guaranteed Pareto optimal solution embedded in the approximation modelling. However, similar to the proposed approach the resource allocation does not fully adapt to the IAB topology, resulting in uneven offloading. Overall, the benchmark results demonstrate that the proposed algorithm achieves near optimal energy efficiency, with only a 5.8% difference from the highest performing approach. As discussed in section 4.1, it also effectively minimises latency across network slices, maintaining QoS according to each slice's specific requirements. From a computational perspective, the proposed algorithm along with the JOBCA and GTCOS algorithm achieve a polynomial time complexity of $O(n * P^2)$, where n denotes the number of tasks and P the maximum path length in the routing DAG. The polynomial complexity ensures scalability with moderate network growth enabling real time operation in dense IAB deployments. In contrast, the PTAS-MAMEC algorithm achieves an exponential time complexity of $O(T^q * (K - q)^3)$, where T is the total number of tasks, K number of simultaneous offloading devices and q is the number of tasks in partial allocation. Despite its high efficiency the exponential growth with respect to network size and approximation precision limits its scalability compared to polynomial time approaches.

5. FUTURE CONSIDERATION AND CONCLUSION

This study proposed a computation offloading framework based on multi-objective optimisation for 5G mmWave IAB-enabled MEC networks, designed to enhance energy efficiency and minimise end-to-end latency while maintaining user quality of service. The results demonstrate that the proposed algorithm achieves a balanced trade-off between energy optimisation and latency reduction across diverse network slices, with polynomial-time complexity that ensures scalability under dense IAB deployments. However, resource allocation remains uneven, particularly for latency critical URLLC tasks, limiting the full potential of offloading. Future work will focus on improving resource allocation fairness across all slices by applying a Pareto front optimisation on collected task data to identify true local minima that maximises energy efficiency, minimises latency, and ensure fair slice offloading results.

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Appendix A: Reflection on Group Work

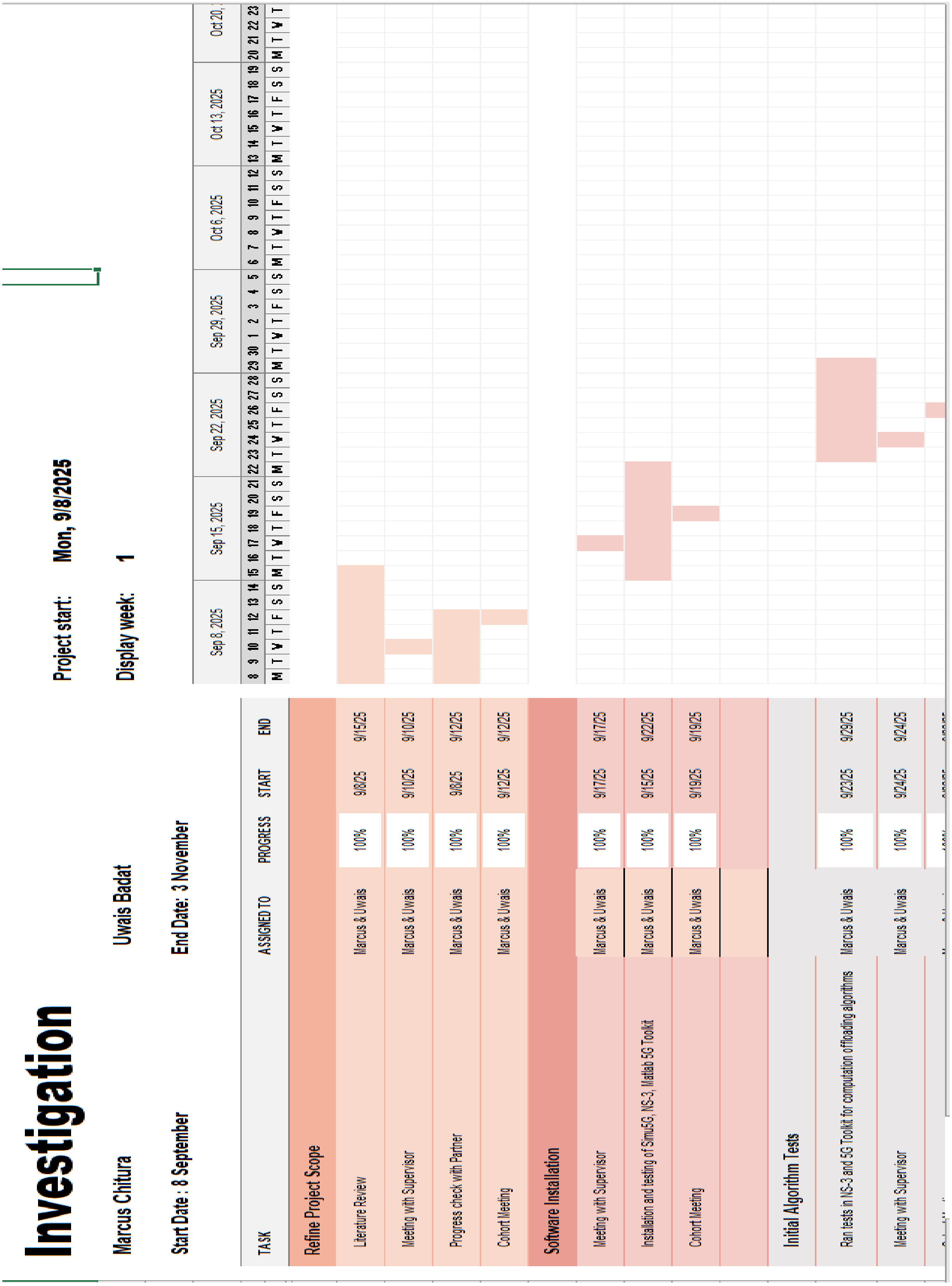

Figure A1: Gnatt Chart of Project Progression

Table A1: Task allocation for project implementation.

Task	Allocated Member	Allocated Member
Literature review & gap identification	Marcus	Uwais
Formulation of multi-objective solution	Marcus	Uwais
Installation & testing of Network Simulator 3 and MATLAB 5G Toolkit	Marcus	Uwais
Setting up base network implementation in NS-3 and 5G Toolkit	Marcus	Uwais
Testing of base network and collecting results for comparison of 4 IAB and gNB setup	Marcus	Uwais
Designing and testing of proposed directed acyclic model	Marcus	
Testing and collecting results for PTAS-MAMEC implementation		Uwais
Testing and collecting results for JOBCA implementation	Marcus	
Testing and collecting results for GTCOS implementation	Marcus	

Table A2: Reflections of investigation project

Aspect	Description	Lesson Learned
Meeting deadlines for results	Several milestones required timely delivery of intermediate results	Improved task management and management of time, along with the estimation of realistic timelines so as to not find self in time crunches.
Communication and coordination	Regular group meetings throughout the project	Strong communication kept us aligned with the project and help solve issues for one another in regards to simulations.
Technical disagreements	Initial differences in selecting the simulation platform and parameter settings for the IAB model.	Consensus was reached through discussion and testing different setups. Learned that early trials and evidence-based comparison reduce conflict and improve decisions.
Scalability of the IAB model	Expanding the model from 4 IAB setup with 10 user equipment to 30 user equipment and a directed acyclic graph increased routing complexity and computation time.	While gradual implementations of project development are the right path, it will introduce problems as the project grows in scope that will cause deviations from projections.

Appendix B: Project Plan

OPTIMISING MULTI ACCESS EDGE COMPUTATION FOR 5G INTEGRATED ACCEESS AND BACKHAUL NETWORK SYSTEMS

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Abstract: This report explores the integration of 5G Integrated Access and Backhaul (IAB) networks with Multi-access Edge Computing (MEC) to reduce mobile user energy consumption. A literature review revealed a research gap in existing offloading strategies, which often neglect the constraints specific to IAB architectures. To address this, the report proposes an optimisation algorithm that performs computation offloading decisions based on estimated delay, device energy profiles, and task priority linked to network slicing. This paper provides details on the IAB topology, the mathematical formulation of the optimisation problem, and the constraints influencing decision making. A comprehensive project management plan supports the implementation, featuring a task allocation table, Gantt chart, and work breakdown structure. This project aims to provide a more practical and energy aware offloading framework that reflects real world limitations of 5G IAB networks.

Key words: 5G, Computationg Offloading, Integrated Access and Backhaul, Multi-access Edge Computing

1. INTRODUCTION

By the late 2010s, the surge in mobile traffic data coupled with a shift toward an integrated digital society in smart cities, smart homes, smart agriculture, industrial automation and autonomous vehicles, drove the development of more reliable network systems. A breakthrough paper by Rappaport *et al* [1], identified millimeter waves (mm Waves), signals above 6GHz as the foundation for fifth generation (5G) mobile communication systems. 5G networks promise key benefits under the three pillars of enhanced mobile broadband (eMBB), ultra-reliable low latency communications (URLLC) and massive machine type communications. Due to the short wavelength and high frequency of mm Waves, they suffer high path loss limiting propagation distance and causing high susceptibility to blockage by buildings, foliage and adverse weather conditions [2].

The standardisation of Integrated Access and Backhaul 5G networks by 3GPP addressed the high path loss of mmWave propagation. IAB enables a 5G base station (gNB) to function as both an access point for user equipment (UE) and a wireless backhaul for other gNBs, reducing reliance on dedicated backhaul infrastructure. Through leveraging mm Waves for backhauling, IAB offers a cost effective alternative to fibre maintaining a high quality of service (QoS) with fast data rates approximately 1-10 Gbps and latency as low as 200 µs [3].

Despite these benefits, IAB networks increase energy consumption for UE, especially at the cell edge or non-line of sight conditions, where uplink power requirements rise. For small or latency sensitive tasks, transmission energy may also exceed the energy saved from local CPU idling. Multi-access edge computing (MEC) offers a solution by enabling cloudlike computation at the edge [4]. By offloading tasks to MEC servers, UE can overcome resource constraints, reduce energy use and achieve better performance. MEC also enhances security, network responsiveness and support for real time applications.

The rest of this paper is structured as follows, section 2 reviews existing offloading strategies and outlines assumptions, constraints and success criteria for the proposed optimisation framework. Section 3 defines the design requirements, and section 4 covers implementation details. Section 5 discusses project management, while section 6 addresses risk mitigation and section 7 concludes the paper.

2. BACKGROUND

2.1. Literature Review

Various computation offloading strategies have been explored in order to optimise performance in 5G networks. A common approach involves the use of joint optimisation theorems [5], [6], [7] which use deterministic multi-variable models for computation offloading schemes. Sardaelitti *et al* [5] for example proposes a computation offloading paradigm set in a muti-cell MEC network. Implemented to optimise radio and computational resources while reducing multi-user energy consumption under latency constraints. In an alternative network Wang *et al* [6], portray joint optimisation using a alternative direction method of multipliers algorithm to optimise content caching strategies for wireless cellular networks in a 5G network. In other avenues of research machine learning based approaches, that are data driven and approximate results offer good results, as shown by Dong *et al* [8] proposal for a deep neural network trained with a digital twin model. This model manages to merge data from the real network for a MEC system, hosting delay and URLLC services achieving 87% energy efficiency for UE. Alternatively, game theory can be applied to simulate multi-agent optimization, as demonstrated by Josilo *et al* [9], who focus on the coordination challenges inherent in computational offloading among autonomous devices or manufacturing machines. Their approach aims to minimize both task completion time and device energy consumption.

The existing methods for computation offloading are implemented in idealised network models, overlooking the structural and bandwidth sharing constraints of IAB networks. Consequently, issues such as backhaul contention, dynamic link quality, and multi-hop propagation delays are often ignored. To address this gap, this project proposes a computation offloading scheduler tailored for mmWave IAB enabled MEC networks. The scheduler makes per task decisions based on estimated end to end delay, device energy profiles and task priority levels mapped to 5G network slices. This optimisation method will use synthetic task models with Poisson distributed arrives to represent random, memoryless traffic along with predefined deadlines and task sizes. This setup aims to reflect real time user demand variability under the practical constraints of IAB networks.

2.2. Assumptions

The assumptions made for the implementation of a MEC server integrated into a 5G IAB network are:

- MEC servers are co-located with donor gNBs or selected IAB nodes.
- Users exhibit low to moderate mobility assuming a urban pedestrian scenario.
- Logical slices are defined by QoS class (e.g., URLLC, eMBB and mMTC enabling task prioritisation.
- Tasks vary in complexity, size and urgency (e.g., artificial reality applications, physics simulations).

2.3. Constraints

The MEC and 5G IAB network integration present the following constraints:

- MEC servers have limited processing power (e.g., cycles per second).
- End to end latency (uplink + compute + downlink) must meet app specific deadlines (e.g., < 1ms for URLLC [10]).
- IAB backhaul links share access spectrum and have limited capacity per slice.
- Slices must not interfere with another slices guaranteed resources and QoS.
- mmWave links are susceptible to blockage, a fallback system for local processing is necessary.

2.4. Success Criteria

The proposed solution for optimising computation offloading is graded on the basis of meeting the following criteria:

- Task completion time is minimised and meets real time deadlines.
- Total energy consumption per task is reduced especially for large and complex tasks (e.g., factory automation, cloud gaming).
- Higher priority slices get preferential access to MEC and backhaul resources.
- System maintains performance as the number of users or network slices increases.
- Higher proportion of tasks are offloaded without violating deadlines.

3. DESIGN REQUIREMENTS

The following are the requirements for the project:

- Design a 5G network with IAB constraints such as limited backhaul, multi-hop limitations, dynamic scheduling and backhaul bottlenecking.
- Implement a custom offloading computation algorithm for the 5G IAB network.
- Conduct network performance tests for latency, throughput, UE energy efficiency, reliability and QoS.
- Mitigate real time traffic issues inherent to 5G IAB communication systems.

4. DESIGN IMPLEMENTATION

This section provides detailed information on the formulation of the optimisation algorithm for computation offloading.

4.1. Network Architecture

The proposed computation offloading architecture in figure 1 consists of a donor gNB with direct fibre backhaul to the core network at a microcell base station (MSB). Multiple IAB nodes connect to the donor gNB via mmWave links, forming the wireless backhaul network system. User equipment is distributed across the coverage area to simulate variable data loads. MEC servers are co-located with the donor gNB and selected IAB nodes to enable task offloading from UE.

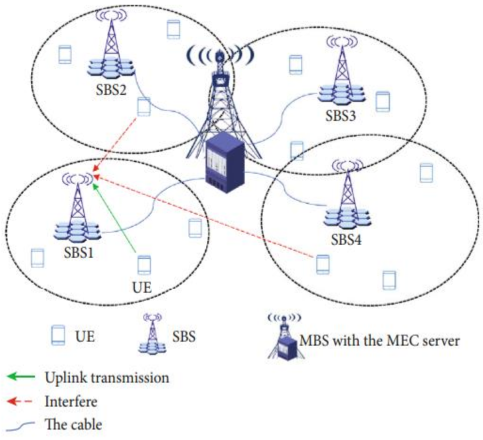

Figure 1: 5G IAB network integrated with a MEC server [11].

4.2. User and Task Model

In the proposed model in figure 1, users generate real time computation tasks according to a Poisson process. Each task has:

- Input data size S_i (0.5 – 200 MB)
- Required CPU cycles C_i (e.g., 1000 cycles per bit)
- Deadline D_i (e.g., 10 – 30 ms)
- Output data size O_i (10 – 50 kB)
- Priority mapped to a network slice – URLLC, URLLC, mMTC

4.3. Mathematical Formulation

This section formalises the proposed computational decision framework for a mmWave IAB enabled MEC network. The model considers key performance trade offs such as task latency, device energy consumption, MEC resource limitation and priority-based scheduling network slicing. Each user periodically generates computational tasks that must either be offloaded to a MEC server or executed locally. The offloading decision for task *i* is defined by a binary variable x_i, where:

$$x_i = \begin{cases} 1 & offload\ tasks\ to\ MEC \\ 0 & execute\ locally \end{cases} \quad (1)$$

This decision variable is central to the optimisation process and governs the execution location of each task. The total latency, T_i^{total} experienced by task *i* in the offloading process is the sum of three components uplink transmission, MEC computation delay and downlink response delay.

$$T_i^{total} = T_i^{uplink} + T_i^{processing} + T_i^{downlink} \quad (2)$$

Each component is defined as follows:

$$T_{uplink} = \frac{S_i}{R_{ul}} \quad (3)$$

where S_i is the task input size in bits and R_{ul} is the uplink transmission rate derived from the Shannon capacity formular:

$$R_{ul} = Blog_2(1 + SINR) \quad (4)$$

where B is the bandwidth and SINR is the uplink signal to interference plus noise ratio. The processing delay is defined by the equation:

$$T_{processing} = \frac{C_i}{f_{MEC}} \quad (5)$$

where C_i is the required number of CPU cycles to complete task *i* and f_{MEC} is the available MEC server processing frequency. The downlink delay is defined by the equation below.

$$T_{downlink} = \frac{O_i}{R_{dl}} \quad (6)$$

where O_i is the output size in bits and R_{dl} is the downlink rate computed similarly to R_{ul}. The objective of the proposed scheme is to minimise the total energy consumption across all UE, while satisfying latency constraints for real time applications. The optimisation problem is presented by following equation.

$$\min_{x_i\{0,1\}} \sum[\, x_i * E_{offload} + (1 - x_i) * E_{local}\,] \quad (7)$$

where $E_{offload}$ is the transmission energy during uplink and E_{local} is the local execution energy estimated using a dynamic voltage and frequency scaling-based model. The offloading energy and local energy are defined by the following equations.

$$E_{offload} = P_{TX} * T_{uplink} \quad (8)$$

where P_{TX} is the transmission power of the UE.

$$E_{local} = \kappa * f_{UE}^2 * C_i \quad (9)$$

where f_{UE} is the UE's CPU frequency and κ is the effective capacitance coefficient representing the device's processing efficiency.

4.4. Constraints of algorithm

Resource limits are implemented using Simu5G's MecAppManager module to configure slice specific resource pools. The constraints of the proposed model are presented in table 1.

Table 1: Proposed constraints for algorithm

Constraint	Expression
Deadline	$T_i^{total} \leq D_i$
MEC CPU limit	$\sum x_i C_i \leq f_{MEC} * t$
Backhaul constraint	1 Gbps limit for mmWave data rate [12].
Slice prioritisation	URLLC > mMTC > eMMB
Offloading benefit	$E_i^{offload} < E_i^{local}$

5. PROJECT MANAGEMENT

The project is scheduled to run from 5 September 2025 to 3 November 2025, with the final report due at the end of this period. A detailed outline of the task allocations in relation to the project is provided in table 2 below.

Table 2: Task allocation for project implementation.

Task	Allocated Member	Allocated Member
Literature review & gap identification	Marcus	Uwais
System architecture design (IAB + MEC)	Marcus	
Simu5G setup and environment configuration		Uwais
Implementation of IAB and MEC algorithm	Marcus	
Simulation, testing and data collection	Marcus	Uwais
Report writing, analysis and submission	Marcus	Uwais

A detailed project timeline is presented in the Gantt chart (see appendix A), additionally, a work breakdown structure (WBS) is included (see appendix B) to decompose the project into smaller, manageable components. The WBS supports effective planning, scheduling, and control by ensuring that all aspects of the project are clearly defined and accounted for. To help ensure the project remains on schedule, weekly meetings with the lecturer will be held every Wednesday, either online via Microsoft Teams or in person if the lecturer is available. Furthermore, internal progress meetings between the team members will occur every Monday, providing a platform to review task progress, address challenges, and maintain accountability for allocated responsibilities.

6. RISK MANAGEMENT

A risk register has been developed (see appendix C) to identify, document, and prioritize potential risks that may impact the project timeline. This register also outlines the mitigation strategies for each identified risk. The most significant risks are associated with tasks exceeding their scheduled deadlines, which could jeopardize the timely completion of the project. Proactive risk monitoring and timely intervention will be key to maintaining the project's schedule and quality.

7. CONCLUSION

This report presented a comprehensive investigation into the integration of 5G integrated access and backhaul (IAB) networks with multi-access edge computing, aimed at addressing the challenge of high energy consumption among mobile users in next-generation networks. Through an in-depth literature review, a clear research gap was identified highlighting the lack of realistic simulation frameworks that account for the constraints of IAB-enabled networks in computation offloading strategies. To address this gap, the report proposed a tailored optimisation algorithm designed to make intelligent offloading decisions based on task deadlines, device energy profiles, and priority mappings to network slices. The system design, including the proposed architecture, mathematical model, and operational constraints, was outlined to guide implementation and evaluation.

Finally, the report included a structured project management plan covering task allocation, a Gantt chart, and a detailed work breakdown structure to ensure effective execution and timely delivery of the project outcomes. The proposed solution sets a foundation for more energy-efficient, delay-aware mobile computing in realistic 5G IAB network environments.

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Appendix C: System Level Parameters

Table C: System Level Parameters for Simulation Model

Parameters	Value
Bandwidth	400 MHz
Carrier Frequency	28 GHz
Pathloss model	UMi Microcell
Thermal noise density	-174 dBm/Hz
Antenna Gain	gNB: 25 dBi, IAB: 25 dBi
Noise Figure	gNB: 5dB, IAB: 5dB, UE: 7dB
Transmit Power (Tx)	gNB: 35 dBm, IAB: 30 dBm, UE: 23 dBm
Antenna Height	gNB: 30m, IAB: 15 – 25m UE: 1.5m
Receiver Power (Rx)	gNB: 5 dB, IAB: 5 dB, UE: 5dB

Appendix D: Slice Specific Data and Latency Values

Table D: Data size and Latency Requirements

Slice Priority	Data Sizes	Latency Requirements
URLLC	Small: 0.005 – 0.02 MB Medium: 0.03 - 0.1 MB Large: 0.15 – 0.3 MB	0.5 – 50 ms
eMMB	Small: 0.1 – 0.5 MB Medium: 1.0 – 5.0 MB Large: 5.0 – 25.0 MB	50 – 200 ms
mMTC	Small: 0.0005 – 0.005 MB Medium: 0.02 - 0.1 MB Large: 0.3 – 2.0 MB	100 – 2000 ms

Appendix E: Parameter Values for Equations

Table E: Parameter Values for Equations 1 - 10

Parameter	Value
CPU Frequency	UE: 1GHz, MEC: 3GHz
Chip energy dependent coefficient	$1 * 10^{-27}$
Spectral efficiency factor	0.7
Processing delay per hop	Uplink: $2ms^{-1,}$ Downlink: $0.2\ ms^{-1}$
Average processing task	$50\ ms^{-1}$
CPU efficiency factor	1 for load < 80 % server task capacity, degrades to 0.8 under full load
UE transmission power	0.5 W

Appendix G: Detailed Pseudocodes

Table G1: Multi-hop Path Selection Algorithm

```
Algorithm 1: Path Selection to Donor
Input: Source IAB node S, DAG topology G
Output: Optimal path P to donor

1: function GET_BEST_PATH_TO_DONOR(S, G)
2:     ALL_PATHS ←
FIND_ALL_PATHS_TO_DONOR (S, 0, G)
3:     BEST_SCORE ← ∞
4:     BEST_PATH ← ∅
5:
6:     for each path ∈ ALL_PATHS do
7:         score ← CALCULATE_PATH_SCORE (path,
G)
8:         if score < BEST_SCORE then
9:             BEST_SCORE ← score
10:            BEST_PATH ← path
11:
12:    return BEST_PATH

13: function CALCULATE_PATH_SCORE (path, G)
14:    score ← LENGTH (path)
15:    for each node ∈ path (excluding donor) do
16:        score ← score + G[node].current Connections
× 0.1
17:    return score
```

Table G2: Link State Management Algorithm

```
Algorithm 2: Link State Assessment
Input: Transmitter Tx, Receiver Rx, Distance D
Output: Link State L, Additional Loss A

function GET_LINK_STATE(Tx, Rx, D)
    key ← (min (Tx, Rx), max (Tx, Rx))

    if link_state_expired(key) then
        los_prob ← CALCULATE_LOS_PROBABILITY(D)
        random_val ← UNIFORM (0,1)

        if random_val ≤ los_prob then
            L ← LOS
            A ← 0 dB
        else if random_val ≤ los_prob + 0.1 then
            L ← BLOCKED
            A ← 50 dB
        else
            L ← NLOS
            A ← 20 dB

        UPDATE_LINK_STATE (key, L, A)

    return L, A
```

Table G3: Offloading Decision Algorithm

```
Algorithm 3: Hybrid Offloading Decision
Input: Task T, Path P, Hop_count H
Output: Offloading decision D

function MAKE_HYBRID_DECISION (T, P, H)

    UL_time, DL_time, worst_SINR ← CALCULATE_MULTIHOP_TIMING (T, P, H)

    // Calculate processing times
    mec_time ← T.cpu_cycles / MEC_frequency
    local_time ← T.cpu_cycles / UE_frequency

    total_offload_time ← UL_time + mec_time + DL_time
    total_local_time ← local_time

    // Energy calculations
    offload energy ← TX power × (UL_time/1000) × (1 + 0.1×H)
    local energy ← κ × UE_freq² × T.cpu_cycles

    // Adaptive weighting based on priority
    if T. priority = 1 then
        α, β ← 0.2, 0.8 // Favor latency
    else if T. priority = 2 then
        α, β ← 0.5, 0.5 // Balanced
    else
        α, β ← 0.7, 0.3 // Favour energy

    // Load-based weight adjustment
    mec_load ← active_tasks / MEC_capacity
    if mec_load > 0.8 then α ← α × 0.7; β ← β × 1.3
    if mec_load < 0.3 then α ← α × 1.3; β ← β × 0.7

    // Cost calculation and decision
    offload_cost ← α × offload_energy + β × total_offload_time
    local_cost ← α × local_energy + β × total_local_time

    if total_offload_time > T. deadline AND total_local_time > T. deadline then
        return "EXECUTE_LOCALLY", "Both exceed deadline"
    else if total_offload_time > T. deadline OR blocked hops > 0 then
        return "EXECUTE_LOCALLY", "Offload infeasible"
    else if local_cost - offload_cost > decision threshold then
        if mec_has_capacity () then
            return "OFFLOAD", "Cost-effective"

    return "EXECUTE_LOCALLY", "Default local execution"
```

Appendix H: Reliability per Network Slice

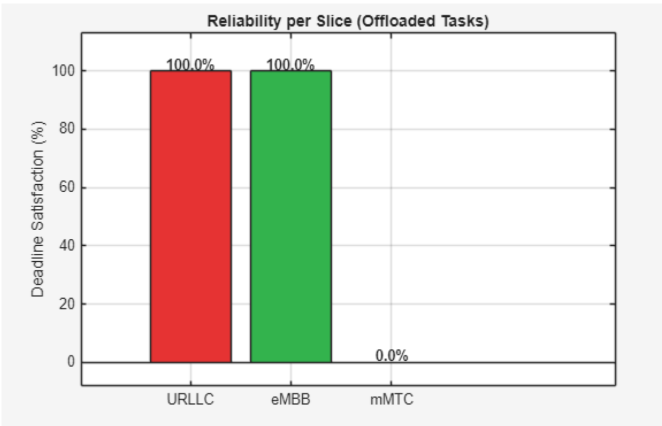

Figure H: Reliability results for the proposed algorithm

Appendix I: Offloading Rates per Algorithm

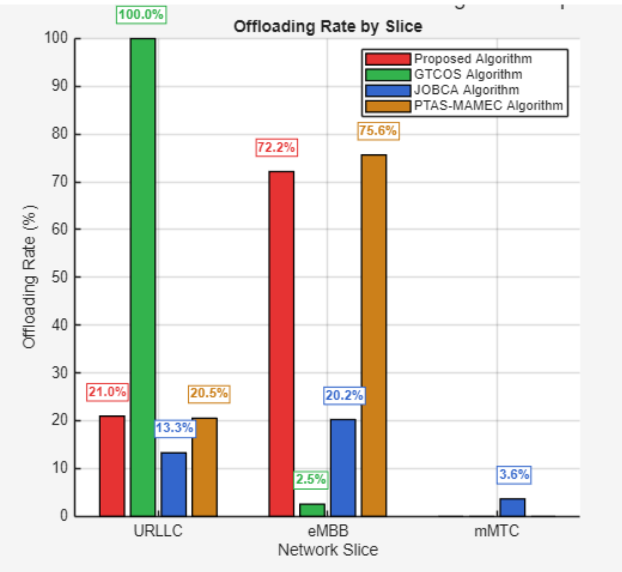

Figure I: Algorithm based offloading rates per network slice